LMS PLATFORM

Infrastructure & Cost Proposal

Prepared for Madushan

February 2026

Confidential — Not for Distribution

EXECUTIVE SUMMARY

This document outlines the 3-phase infrastructure plan and associated third-party service costs for building an online Learning Management System (LMS). Development costs are excluded.

1. Project Overview

We are building a full-featured LMS platform from the ground up using the MERN stack (MongoDB, Express, React, Node.js). The platform will support instructors creating and uploading courses, students enrolling and watching video content, and administrators managing the platform.

The project is divided into 3 phases:

- Phase 1 — MVP: Core platform with manual payment & email notifications
- Phase 2 — Growth: Online payment integration, video streaming upgrade, progress tracking
- Phase 3 — Scale: Advanced features, analytics, mobile app, moderation tools

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2. Phase 1 — MVP

Scope

- User authentication (Sign up, Login, Google OAuth)
- Instructor dashboard — create & upload courses
- Student dashboard — browse & request enrollment
- Admin panel — approve/reject enrollments after offline payment
- Manual payment flow (admin confirms payment offline)
- Email notifications (welcome, enrollment confirmation, password reset, new course)
- Basic search and category filtering
- Video hosting via Cloudinary (free tier)

LearnHub Categories Search for anything Teach on LearnHub

Home / Development / Web Development / React

Complete React Developer Bootcamp 2024

Become a Senior React Developer! Build massive projects, master React, Redux, Node.js, and more from scratch.

Bestseller 4.8 ★★★★★ (2,400 ratings) • 12,500 students

Created by Dr. Angela Yu Last updated 11/2024 English CC

Overview Curriculum Instructor Reviews

What you'll learn

- ✓ Build powerful, fast, user-friendly and reactive web apps
- ✓ Apply for high-paid jobs or work as a freelancer in one of the most demanded sectors
- ✓ Master Redux for state management in large scale applications
- ✓ Provide amazing user experiences by leveraging the power of JavaScript with ease
- ✓ Learn all about React Hooks and React Components
- ✓ Become fluent in the toolchain of modern React developers

Course Content

15 sections • 145 lectures • 24h 10m total length Expand all sections

Introduction to React 3 lectures • 12min

- Welcome to the Course Preview 04:21
- What is React? 03:45
- Setting up the Environment 05:10

Javascript Refresher 5 lectures • 45min

React Basics & Components 12 lectures • 1h 20min

Instructor

Dr. Angela Yu

Developer and Lead Instructor

- 4.7 Instructor Rating
- 150,000 Reviews
- 500,000 Students
- 7 Courses

I'm Angela, I'm a developer with a passion for teaching. I'm the lead instructor at the London App Brewery, London's leading Programming Bootcamp. I've helped hundreds of thousands of students learn to code and change their lives by becoming a developer.

Student Feedback

4.8 ★★★★½ Course Rating

- ★ 57%
- ★ 32%
- ★ 8%
- ★ 2%
- ★ 1%

JD **John Doe** ★★★★★

This course changed my career. I went from knowing zero React to building full-stack applications. Highly recommended!

SM **Sarah M.** ★★★★★

Great explanations and pacing. The section on Redux was a bit fast, but re-watching helped. Overall excellent value.

$49.99 ~~$94.99~~ 47% off

Request Enrollment

Add to Wishlist

This course includes:

- 45 hours on-demand video
- 50 Coding Exercises
- 12 Articles
- Full lifetime access
- Access on mobile and TV
- Certificate of completion

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Alex Morgan
Super Admin

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Dashboard Overview

Welcome back, here's what's happening with your platform today.

Search...

Total Users	Total Courses	Pending Enrollments	Revenue
12,450	340	45	$45,200
+5% vs last month	+12% vs last month	Action Needed · Since yesterday	+3% vs last month

Pending Enrollment Requests

View All

STUDENT	COURSE	DATE	ACTIONS
Alice Johnson alice.j@example.com	UX Design Fundamentals	Oct 24, 2023	
Bob Smith bob.s@example.com	Python for Beginners	Oct 23, 2023	
Diana Evans diana.e@example.com	Digital Marketing 101	Oct 21, 2023	
Ethan Hunt ethan.h@example.com	Advanced React Patterns	Oct 20, 2023	

New Registrations

- SJ Sarah Jenkins — Just now — Student
- Michael Chen — 2 hours ago — Instructor
- EW Emily Wong — 5 hours ago — Student
- David Kim — Yesterday — Student

View All Users

Recent Courses

Newly published content on the platform

Filter

COURSE TITLE	CATEGORY	ENROLLED STUDENTS	INSTRUCTOR	STATUS
Mastering Figma Constraints	Design	1,204	Sarah Miller	Active

Filters

Clear All

CATEGORY

- [x] Development 120
- [ ] Design 45
- [ ] Business 32
- [ ] Marketing 18

LEVEL

- [ ] Beginner
- [ ] Intermediate
- [ ] Advanced

RATING

- [ ] ★★★★★ & Up
- [ ] ★★★★★ & Up

PRICE

- [ ] Paid
- [ ] Free

Explore Courses

Search results...

Most Popular

Development × Beginner ×

DEVELOPMENT 4.8 ★ (1.2k)

The Complete 2024 Web Development Bootcamp

Dr. Angela Yu

62h total • All Levels **$19.99**

BUSINESS 4.6 ★ (850)

Product Management 101: From Idea to Launch

Michael Chen

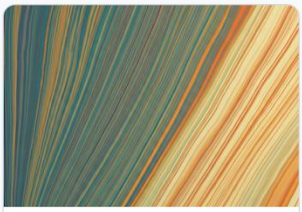
12h total • Beginner **$24.99**

DEVELOPMENT 4.9 ★ (230)

Advanced React Patterns and Performance

Sarah Smith

8h total • Advanced **$39.99**



DESIGN 4.7 ★ (1.8k)

Graphic Design Masterclass - Learn GRE...

Lindsay Marsh

28h total • Beginner **$14.99**

DATA SCIENCE 4.5 ★ (560)

Python for Data Science and Machine Learning

Jose Portilla

45h total • Intermediate **$12.99**

MARKETING 4.8 ★ (3.2k)

The Complete Digital Marketing Course - 12 in 1

Rob Percival

22h total • Beginner **$18.99**

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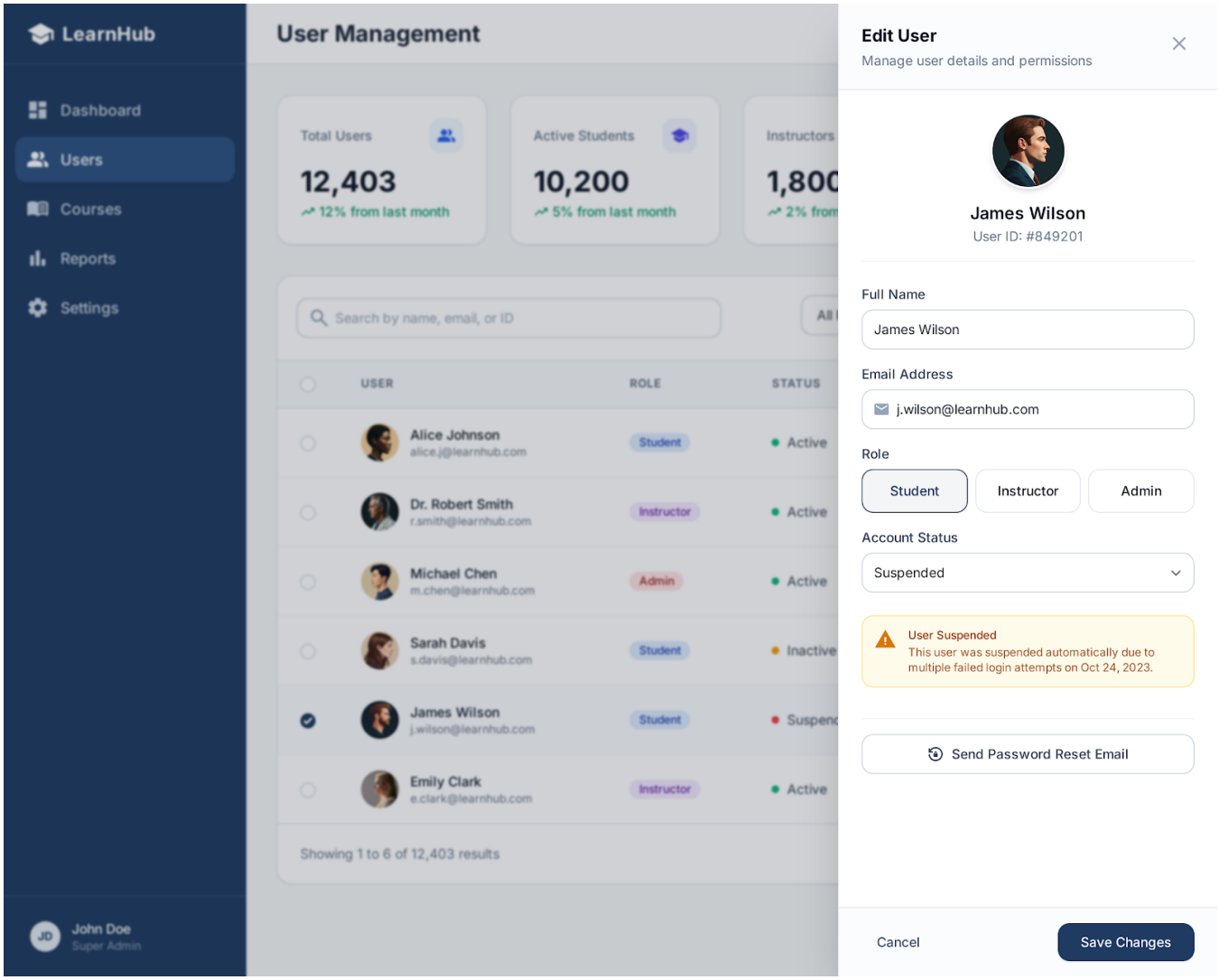
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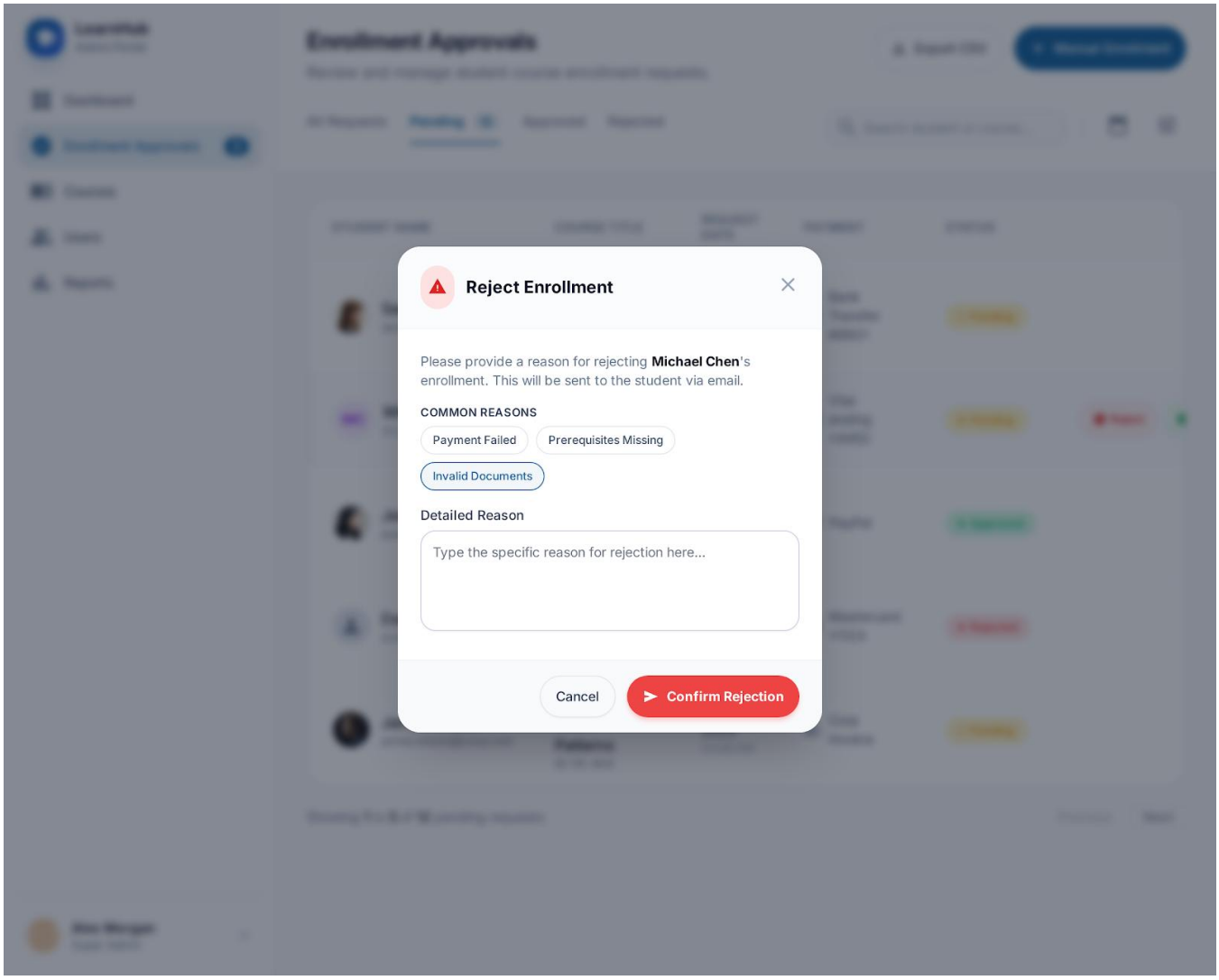
Dashboard
Users
Courses
Reports
Settings
John Doe
Super Admin
User Management
Total Users
12,403
12% from last month
Active Students
10,200
5% from last month
Instructors
1,800
2% from
Search by name, email, or ID
All
USER
ROLE
STATUS
Alice Johnson
alice.j@learnhub.com
Student
Active
Dr. Robert Smith
r.smith@learnhub.com
Instructor
Active
Michael Chen
m.chen@learnhub.com
Admin
Active
Sarah Davis
s.davis@learnhub.com
Student
Inactive
James Wilson
j.wilson@learnhub.com
Student
Suspend
Emily Clark
e.clark@learnhub.com
Instructor
Active
Showing 1 to 6 of 12,403 results
Edit User
Manage user details and permissions
James Wilson
User ID: #849201
Full Name
James Wilson
Email Address
j.wilson@learnhub.com
Role
Student
Instructor
Admin
Account Status
Suspended
User Suspended
This user was suspended automatically due to multiple failed login attempts on Oct 24, 2023.
Send Password Reset Email
Cancel
Save Changes


Reject Enrollment
Please provide a reason for rejecting Michael Chen's enrollment. This will be sent to the student via email.
COMMON REASONS
Payment Failed
Prerequisites Missing
Invalid Documents
Detailed Reason
Type the specific reason for rejection here...
Cancel
Confirm Rejection

LearnHub

Search courses...

Sarah Jenkins
Student

Hi, Sarah! 👋

Let's learn something new today.

ENROLLED
5

COMPLETED
2

IN PROGRESS
3

All Courses | In Progress | Completed | Pending Approval

★ 4.9

DESIGN

Advanced UX Design Principles

By Jane Doe

Progress 75%

Continue Learning →

DEVELOPMENT

React for Beginners

By John Smith

Progress 30%

Continue Learning →

Pending Approval

DATA SCIENCE

Data Science Fundamentals

By Alice Chen

Not Started

Awaiting Approval

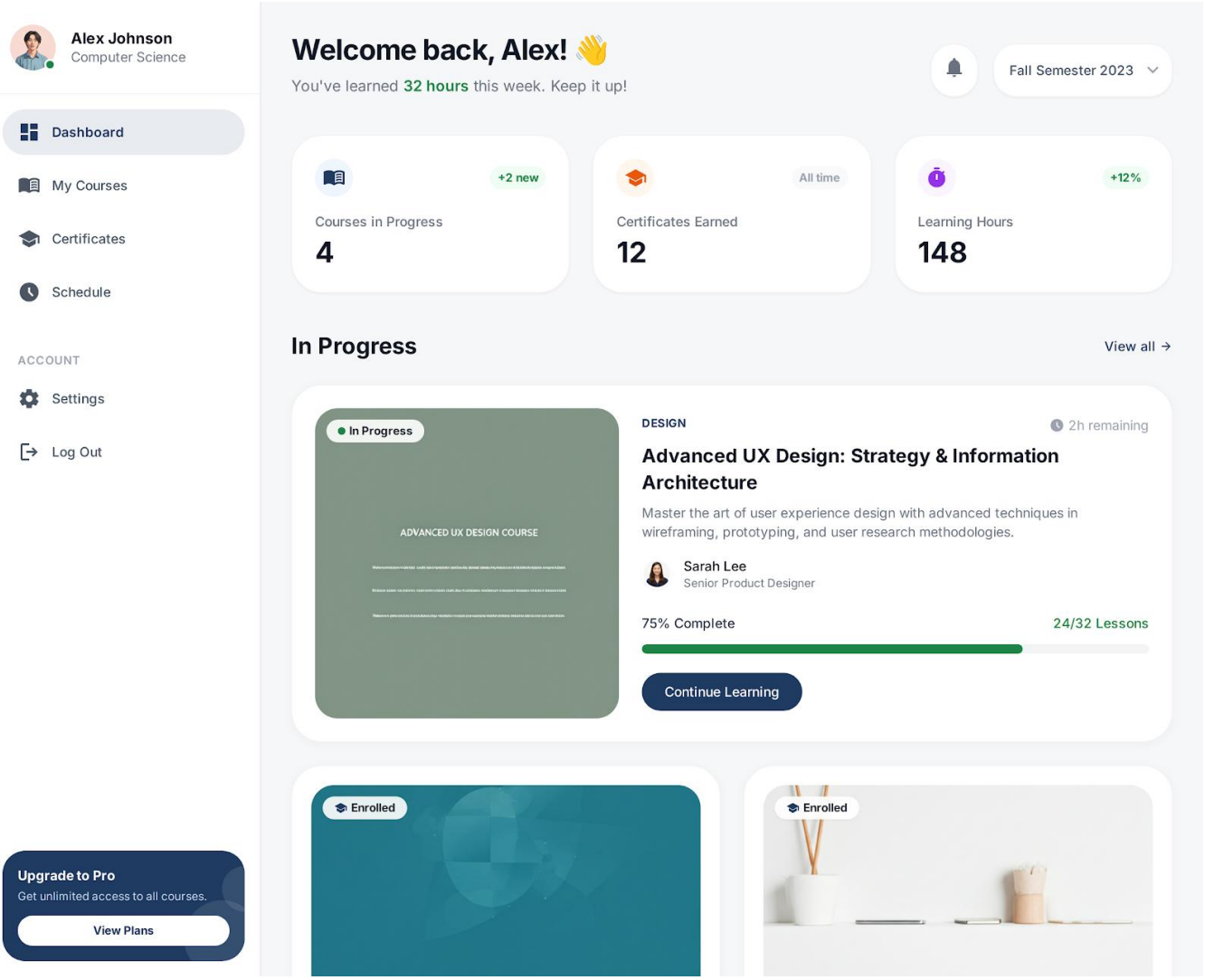
MARKETING

Digital Marketing Mastery

By David Miller

Progress 10%

Continue Learning →


Alex Johnson
Computer Science
Dashboard
My Courses
Certificates
Schedule
ACCOUNT
Settings
Log Out
Upgrade to Pro
Get unlimited access to all courses.
View Plans
Welcome back, Alex! 👋
You've learned 32 hours this week. Keep it up!
Fall Semester 2023
+2 new
Courses in Progress
4
All time
Certificates Earned
12
+12%
Learning Hours
148
In Progress
View all →
In Progress
ADVANCED UX DESIGN COURSE
DESIGN
2h remaining
Advanced UX Design: Strategy & Information Architecture
Master the art of user experience design with advanced techniques in wireframing, prototyping, and user research methodologies.
Sarah Lee
Senior Product Designer
75% Complete
24/32 Lessons
Continue Learning
Enrolled
Enrolled

LMS Platform

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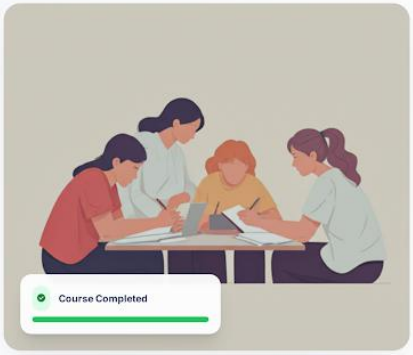

25k+	800+	120	4.9
Active Students	Expert Mentors	Countries	Average Rating

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4.8 ★ (12,345)

Complete Python Bootcamp: From Zero to...

Jose Portilla

~~$89.99~~ $19.99

4.6 ★ (8,900)

The Ultimate Drawing Course - Beginner to...

Jaysen Batchelor

~~$109.99~~ $24.99

4.9 ★ (2,100)

MBA in a Box: Business Lessons from a CEO

365 Careers

~~$49.99~~ $14.99

4.7 ★ (5,430)

Digital Marketing Masterclass - 23 Courses...

Phil Ebiner

~~$199.99~~ $34.99

Top Categories

Development Design Business Marketing Photography Music

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alex.doe@company

Invalid email address

Password

Enter your password

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Instructor Dashboard

Tuesday, February 17, 2026

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- Earnings
- Settings

John Smith
Senior Instructor

Welcome back, John! 👋

Here's what's happening with your courses today.

Total Courses	Total Students	Pending Approvals	Total Revenue
12	1,240	5	$12,450
+2%	+15%	New	+8%

My Courses

Manage your active courses and drafts.

+ Add New Course

Course Title	Category	Status	Students	Act
Advanced React Patterns	Development	Published	842	
Introduction to UX Design	Design	Draft	-	
Data Science Fundamentals	Data Science	Published	398	
Cybersecurity Basics	IT & Security	Reviewing	-	

View All Courses

Recent Enrollment Requests

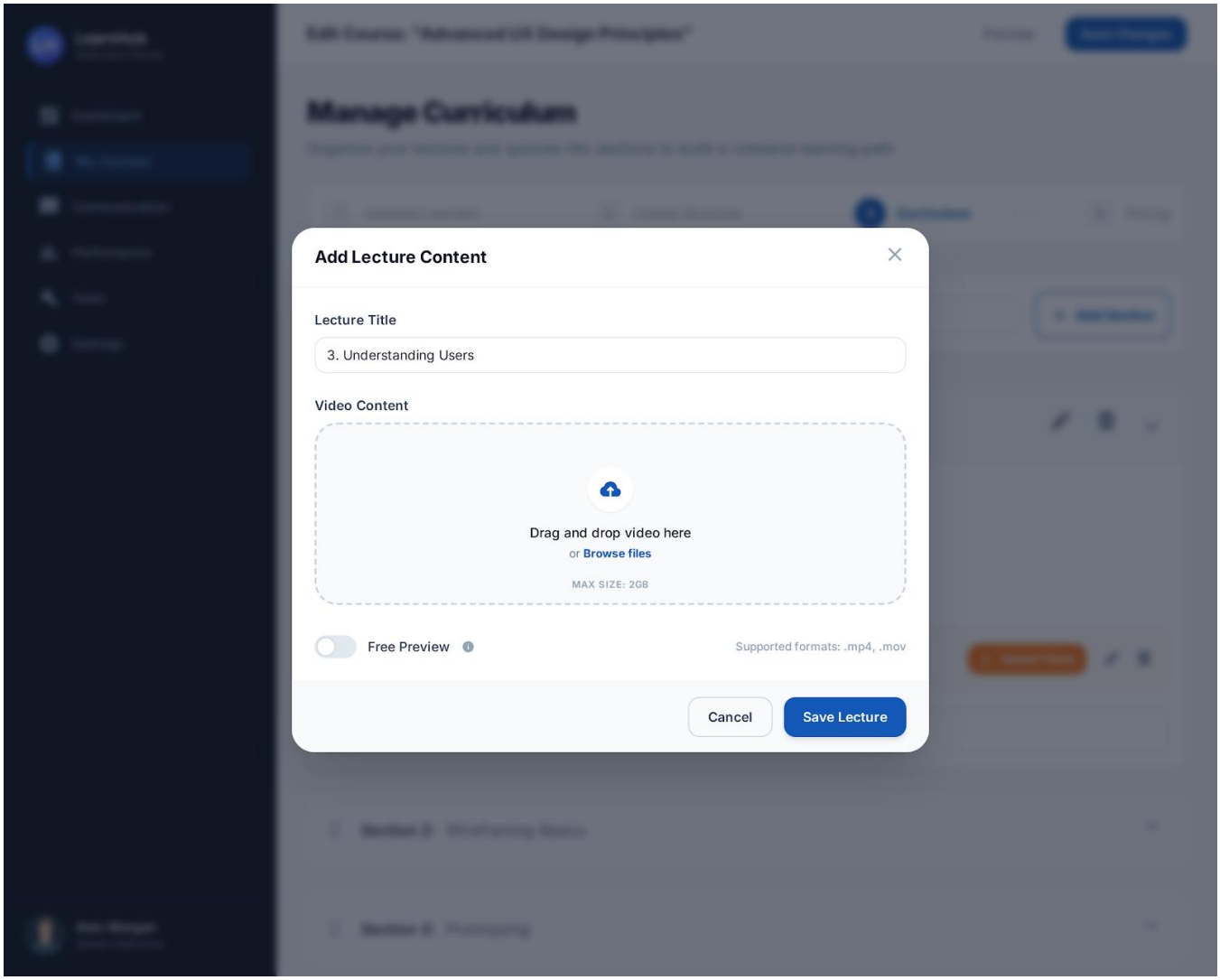
Review student access requests.

Sarah Connor — Advanced React Patt... — 2m ago
Approve | Reject

Michael Chen — Data Science Funda... — 15m ago
Approve | Reject

Jessica Day — Introduction to UX De... — 1h ago
Approve | Reject

View All Requests


Manage Curriculum
Add Lecture Content
Lecture Title
3. Understanding Users
Video Content
Drag and drop video here
or Browse files
MAX SIZE: 2GB
Free Preview
Supported formats: .mp4, .mov
Cancel
Save Lecture

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- **Marketing** 650+ Courses
- **Photography** 400+ Courses
- **Data Science** 1,100+ Courses
- **Music** 320+ Courses
- **Personal Dev** 1,500+ Courses

Featured Courses

Hand-picked by our experts for your career growth.

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4.8 (2.5k reviews)

The Complete Python Bootcamp: From Zero to...

J. Smith $14.99

Marketing

4.9 (1.1k reviews)

Digital Marketing Masterclass 2024

S. Lee $19.99

Design

4.7 (850 reviews)

UI/UX Design Specialization

M. Johnson $24.99

Business

4.6 (3.2k reviews)

MBA in a Box: Business Lessons

D. Brown $12.99

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Login Sign Up

Check your email

We have sent password recovery instructions to your email **john@example.com**.

Back to Login

Didn't receive the email?

Click to resend

Privacy Terms Help

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Contact Support

Forgot password?

No worries, we'll send you reset instructions.

Email address

name@company.com

Reset password

Back to log in

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LH

Courses > Create New

Web Development Masterclass

Preview | Save Draft

Plan ✓ — Media ✓ — 3 Curriculum — 4 Publish

Curriculum Builder

Organize your course into sections and lectures.

Bulk Upload

Section 1: Introduction

- Welcome to the course
  2:30 • Content ready
- Course Overview & Resources
  3 files

+ Add Lecture

Section 2: Setup — 0 Lectures

Add New Section

Quick Tips

- Keep lectures under **10 minutes** for better student engagement.
- Start every section with a brief **intro video** to set expectations.
- Mix video content with quizzes or assignments to test knowledge.

Read full guide ↗

Course Content

Summary of your progress

2	2
Sections	Lectures

15 min
Total Duration

← Previous: Media

Next: Publish →

Create New Course

Draft in progress

Save Draft Exit

1 Basic Info — 2 Media — 3 Curriculum — 4 Publish

Course Media

Add visual assets to make your course stand out on the marketplace.

Course Thumbnail

Upload your course image here. Important guidelines: 750×422 pixels; .jpg, .jpeg, .gif, or .png. no text on image.

Promotional Video

RECOMMENDED

Students who watch a well-made promo video are 5x more likely to enroll.

← Previous

Next: Curriculum →

Create New Course

Step 1 of 4 • Basic Information

Save Draft

Next: Media →

1 Basic Info — 2 Media — 3 Curriculum — 4 Publish

Course Details

Course Title *

e.g., Advanced React Patterns for Senior Developers

Short Description 0/120

A brief summary of your course.

Full Description

B I

Enter a detailed description of your course...

Categorization & Pricing

Category

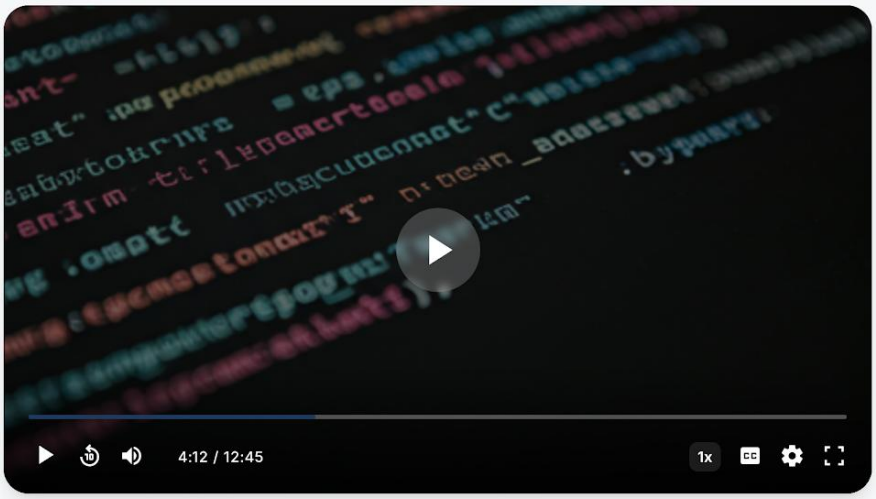
Select a category

Level

Select difficulty

Language

Price (USD)



React Hooks — useState and useEffect Deep Dive

Last updated 2 days ago

Mark Complete

Overview | Resources 3 | Notes | Discussion

About this lesson

In this comprehensive deep dive, we will explore the two most fundamental hooks in React: `useState` and `useEffect`. Understanding these hooks is crucial for managing state and side effects in functional components.

We'll start by refactoring a class-based component to a functional one, highlighting the simplifications hooks provide. Then, we'll tackle common pitfalls like infinite loops in dependency arrays and stale closures.

What you'll learn

- Managing local component state with useState
- Handling side effects and lifecycle events with useEffect
- Cleanup functions and avoiding memory leaks

Pro Tip: Always double-check your dependency array in useEffect to ensure your effect runs exactly when intended.

Course Progress 35%

12 of 48 lessons completed

Curriculum Expand All

Section 1: Introduction
4 Lessons • 25m

Section 2: React Basics
6 Lessons • 1h 15m

1. Introduction to JSX
8:45

2. Components and Props
12:30

3. React Hooks — useState & useEffect
15:20 NOW

4. Custom Hooks
10:15

5. Handling Events
09:40

Section 3: Advanced Patterns
5 Lessons • 55m

Section 4: Performance
3 Lessons • 45m

Course Materials
Download source code & slides

All Courses

Showing 48 results for "Design"

Filters

Clear All

Categories

- [x] Development (120)
- [ ] Business (45)
- [ ] IT & Software (32)
- [ ] Design (88)

Level

- [ ] Beginner
- [x] Intermediate
- [ ] Expert

Price $10 - $200

$0 $500+

Apply Filters

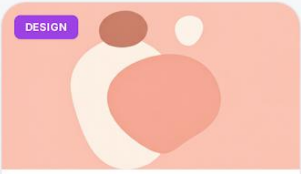
Sort by: Most Popular

★ 4.8 (2.4k) $14.99

Complete Python Bootcamp: Go from zero to hero in Python 3

Master Python by building 100 projects in 100 days. Learn data...

Jose Portilla 22h 155



★ 4.9 (5k) Free

UI/UX Design Masterclass: From Beginner to Pro

Learn the complete design process from user research to high fidelity...

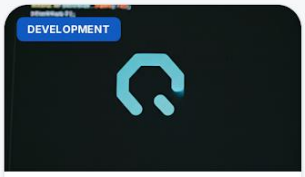
Sarah Doe 12h 42

★ 4.5 (890) $89.99

MBA in a Box: Business Lessons from a CEO

Business Strategy, Management, Marketing, Accounting, Decision...

David Green 8h 24



★ 4.7 (1.2k) $12.99

React - The Complete Guide 2024

Dive in and learn React.js from scratch! Learn Reactjs, Hooks,...

Maximilian S. 48h 240

★ 4.4 (320) $24.99

Digital Marketing Agency: SMMA

Learn how to build a successful social media marketing agency fro...

Robin & Jes... 18h 95

★ 4.6 (750) $19.99

The Complete Cyber Security Course

Volume 1: Hackers Exposed. Join this comprehensive course to master...

Nathan House 11h 55

‹ 1 2 3 ... 10 ›

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Phase 1 — Service Cost Breakdown

Service	Free Tier	Monthly Cost	Notes
Cloudinary (Video)	25GB storage + 25GB bandwidth	**$0 / month**	Video hosting & delivery
MongoDB Atlas	512MB database	**$0 / month**	Database hosting
Vercel (Frontend)	Unlimited hobby	**$0 / month**	React frontend hosting
Render (Backend)	Free tier (sleeps)	**$0 – $7 / month**	$7/mo recommended to avoid sleep
SendGrid (Email)	100 emails/day	**$0 / month**	Transactional email
Domain Name	—	**~$1 / month**	~$12/year (optional for MVP)
PHASE 1 TOTAL		**$0 – $8 / month**	**Effectively FREE to launch**

Note: Cloudinary free tier supports approximately 17–35 hours of watched content per month. Video files should be limited to 720p during MVP to maximize free tier usage.

3. Phase 2 — Core LMS + Payment Integration

Scope

- Stripe payment integration (replace manual flow)
- Instructor revenue split & payouts via Stripe Connect
- Video streaming upgrade: AWS S3 + CloudFront OR Mux
- Course progress tracking per student
- Quizzes & completion certificates (PDF generation)
- Reviews and ratings system
- Coupon and discount code system
- Advanced email sequences (progress reminders, promotions)

Phase 2 — Service Cost Breakdown

Service	Plan	Monthly Cost	Notes
Mux (Video Streaming)	Pay per use	**~$20–$80 / month**	$0.015/min stored + $0.0025/min delivered. Scales with usage.
OR: AWS S3 + CloudFront	Pay per use	**~$15–$60 / month**	More control, more setup. Scales well.
Stripe Payments	Per transaction	**2.9% + $0.30 / txn**	No monthly fee. Deducted from revenue.
Stripe Connect (Payouts)	Per payout	**0.25% per payout**	For instructor revenue splitting
MongoDB Atlas	M10 Cluster	**$57 / month**	Upgrade needed for production scale
Render / Railway (Backend)	Starter / Pro	**$20–$25 / month**	Always-on server with more resources
SendGrid	Essentials	**$19.95 / month**	50,000 emails/month
Vercel (Frontend)	Pro	**$20 / month**	Teams, analytics & more bandwidth
PHASE 2 TOTAL (est.)		**$155–$220 / month**	**Excludes Stripe txn fees (from revenue)**

Important: Stripe transaction fees (2.9% + $0.30) are deducted from course sales revenue, not a fixed monthly overhead. At $10,000/month in sales, Stripe fees are approximately $320.

4. Phase 3 — Scale & Growth

Scope

- Mobile application (React Native — iOS & Android)
- Advanced search powered by Algolia or Elasticsearch
- Recommendation engine for personalized learning
- Admin content moderation panel
- Analytics dashboard for instructors & admins
- Multi-language & localization support
- Live sessions / webinar integration
- Corporate/team accounts (B2B LMS)

Phase 3 — Service Cost Breakdown

Service	Plan	Monthly Cost	Notes
All Phase 2 Services	Upgraded tiers	**~$220 / month**	Baseline carried forward from Phase 2
Algolia (Search)	Grow Plan	**$50–$100 / month**	Fast, scalable full-text search
AWS (Expanded Infra)	EC2 + Lambda + RDS	**$100–$300 / month**	Depends heavily on traffic & usage
Twilio / Daily.co (Live)	Pay per use	**$50–$200 / month**	For live sessions / webinars (optional)
Apple Dev Account	Annual	**~$8 / month**	$99/year for iOS app distribution
Google Play Account	One-time	**$0 (after $25 setup)**	$25 one-time fee for Android distribution
Monitoring (Datadog / Sentry)	Team Plan	**$50–$80 / month**	Error tracking, performance monitoring
PHASE 3 TOTAL (est.)		**$500–$900+ / month**	**Scales with platform traffic & users**

Note: Phase 3 costs are highly variable. Live sessions and AWS infrastructure scale directly with platform usage and active users. These are estimates based on a platform serving 500–2,000 active monthly users.

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5. Cost Summary — All Phases

The table below provides a high-level cost comparison across all three phases. These are infrastructure and third-party service costs only and exclude all development fees.

Phase	Timeline	Key Services	Monthly Cost	Status
Phase 1 — MVP	Month 1–3	Cloudinary, MongoDB, Render, SendGrid	**$0 – $8**	**Ready to Launch**
Phase 2 — Core LMS	Month 4–8	Mux/AWS, Stripe, MongoDB M10, SendGrid	**$155 – $220**	**Post-Validation**
Phase 3 — Scale	Month 9+	AWS, Algolia, Mobile, Live Sessions	**$500 – $900+**	**Growth Stage**

Key Financial Principles

1. **Start free, scale with revenue —** MVP costs nothing. Only upgrade services when paying users are onboarded.
2. **Stripe fees are not overhead —** Payment processing fees are deducted from revenue, not fixed costs.
3. **Phase 3 costs are usage-based —** Higher traffic = higher cost, but also higher revenue. Profitability improves with scale.
4. **Migration path is clean —** Services are designed to be swapped (e.g., Cloudinary → Mux) with minimal frontend changes.

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6. Technology Stack Reference

The following technologies are used to build the platform across all phases:

Layer	Technology	Purpose
Frontend	React.js + Vite + Tailwind CSS	Student, Instructor, and Admin UI
Backend	Node.js + Express.js	REST API, business logic, auth
Database	MongoDB + Mongoose	Users, courses, enrollments, transactions
Authentication	JWT + Google OAuth (Passport.js)	Secure user authentication
Video (MVP)	Cloudinary	Free video hosting and delivery
Video (Phase 2)	Mux or AWS S3 + CloudFront	HLS streaming, adaptive quality, DRM
Payments	Stripe + Stripe Connect	Course purchases, instructor payouts
Email	Nodemailer + SendGrid	Transactional emails, notifications
Search	MongoDB Text Index → Algolia (P3)	Course search and filtering
Hosting	Vercel (FE) + Render/Railway (BE)	Scalable cloud deployment

Disclaimer

All costs listed in this document are estimates based on publicly available pricing as of February 2026. Actual costs may vary based on usage, number of users, video hours consumed, and third-party pricing changes. Development costs, custom tooling, and internal team resources are excluded from this proposal.